AI-Based Research Paper Analysis Using Transformer Models **Abstract** The increasing number of scientific publications has made it difficult for researchers to stay updated with recent developments. Artificial Intelligence (AI), particularly transformer-based language models, offers a solution by automating tasks such as summarization, topic extraction, and research gap identification. This study evaluates the effectiveness of AI models in analyzing research papers and assisting researchers during literature reviews. **1. Introduction** Research output has grown rapidly across all academic disciplines. Manually reviewing large numbers of papers is time-consuming and often inefficient. AI-powered language models can process large volumes of text and generate useful insights, helping researchers save time and improve productivity. **2. Methodology** A dataset of 500 research papers from computer science and engineering domains was analyzed. Transformer-based language models were used to perform three tasks: * Research paper summarization * Topic extraction * Research gap identification The generated outputs were evaluated by experts based on accuracy, relevance, coherence, and usefulness. **3. Results** The AI models performed well in summarization tasks, achieving an average accuracy of over 85%. Generated summaries successfully captured key objectives, methodologies, and findings of research papers. Topic extraction also showed strong performance, with models accurately identifying major themes such as machine learning, cybersecurity, and data analytics. The average F1-score was approximately 0.88. Research gap identification was more challenging. While AI systems suggested meaningful future research directions, expert review was still necessary to validate the recommendations. **4. Discussion** The results demonstrate that transformer-based models can significantly reduce the effort required for literature reviews. Researchers can quickly understand large collections of papers without reading every document in detail. However, AI-generated outputs occasionally contained inaccuracies or missed important contextual information. Therefore, these systems should be viewed as research assistants rather than replacements for human expertise. Computational requirements also remain a limitation, as larger models require significant hardware resources. **5. Applications** AI-powered research analysis tools can support: * Academic literature reviews * Student research projects * Industrial technology monitoring * Scientific knowledge management * Research trend analysis Such systems can improve efficiency and help researchers discover relevant information more quickly. **6. Conclusion** Transformer-based AI models provide an effective solution for automated research paper analysis. They demonstrate strong performance in summarization and topic extraction while offering useful support for identifying research opportunities. Although human oversight remains necessary, AI-assisted research tools have the potential to transform how

researchers interact with scientific literature. **Keywords:** Artificial Intelligence, Natural Language Processing, Research Analysis, Literature Review, Transformer Models.